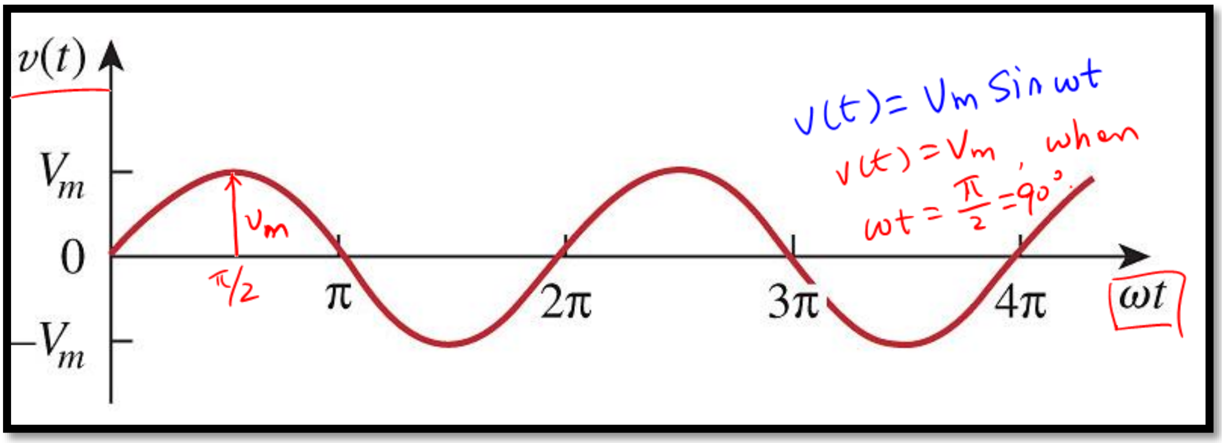
L#28: Sinusoids

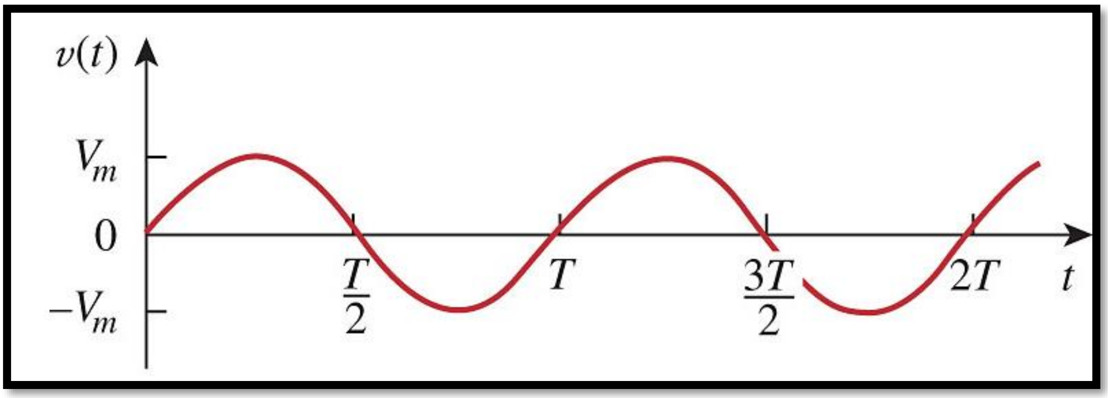
A sinusoid is a signal has the ____sine______ or ____cosine______ functions.



$v(t) = V_m \sin(\omega t + \phi)$	
V_m:	amplitude of a sinusoid signal,
ω:	angular speed [rad/s]
t:	time, [s] $\quad \underset{[rad/s]}{\omega} \cdot \underset{[s]}{t}$.
ωt:	argument of a sine or cosine function. rad
ϕ:	angle [rad], [degree]

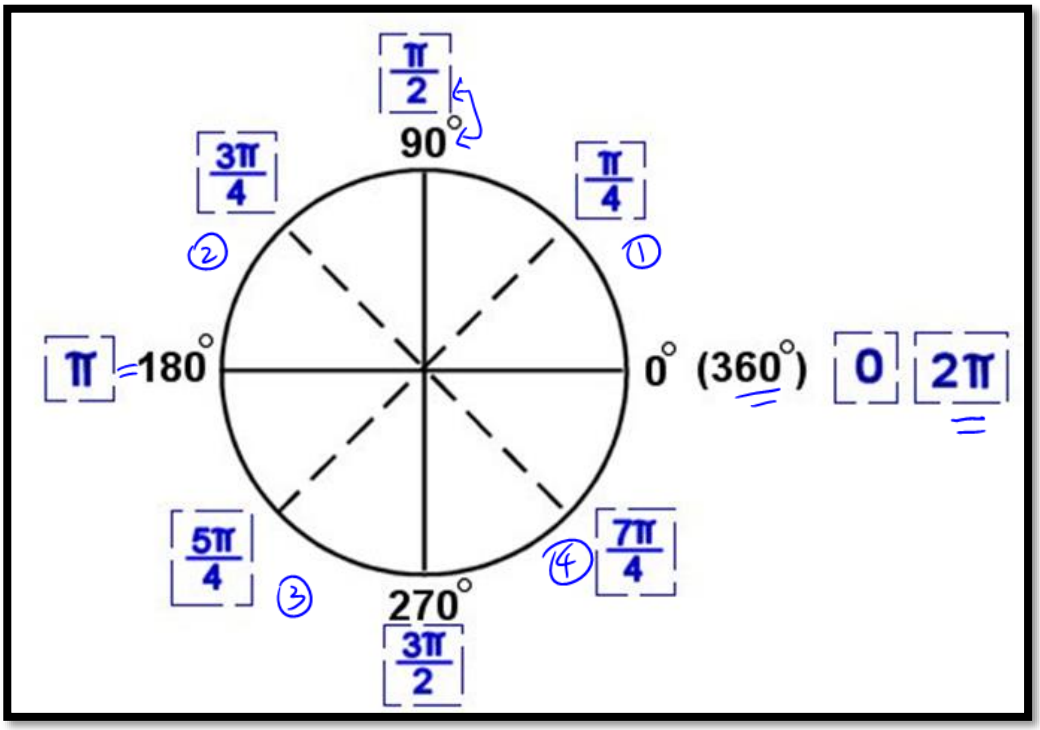
$\pi/3 \qquad 45°$.

Sinusoids vs. time (not ωt)



T=	period $\omega T = 2\pi$, $\Rightarrow T = \frac{2\pi}{\omega}$, $f(t) = \sin \omega t$, $f(T) = \sin \omega T$, $t = T$ $= \sin \omega \cdot \frac{2\pi}{\omega} = \sin 2\pi = 0$.
at t=T:	$f(t=T) = 0$. $T = \frac{2\pi}{\omega}$
f:	frequency of a sinusoid signal, [Hz]. inverse of T : $f = \frac{1}{T}$, $T = \frac{2\pi}{\omega}$ $\Rightarrow f = \frac{\omega}{2\pi}$
ω	$\omega = 2\pi \cdot f$ [rad].

Radians & Degree



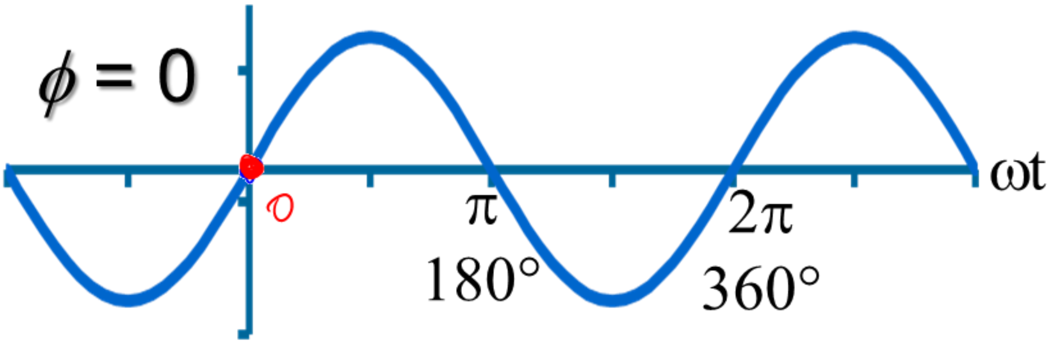
	$360° = \underline{\ 2\pi\ }$ **rad**
	$180° = \underline{\ \pi\ }$ **rad**
θ φ **deg→rad:**	$\varphi = \frac{\theta}{180°} \cdot \pi$, $\frac{60°}{180°} \cdot \pi = \frac{\pi}{3}$.
rad→deg:	$\theta = \frac{\varphi}{\pi} \cdot 180°$, $\frac{\pi/4}{\pi} \cdot 180° = 45°$.

Phase angle

$$v(t) = V_m \sin(\omega t + \phi)$$

$\omega t + \phi = 0 \Rightarrow v(t) = 0.$

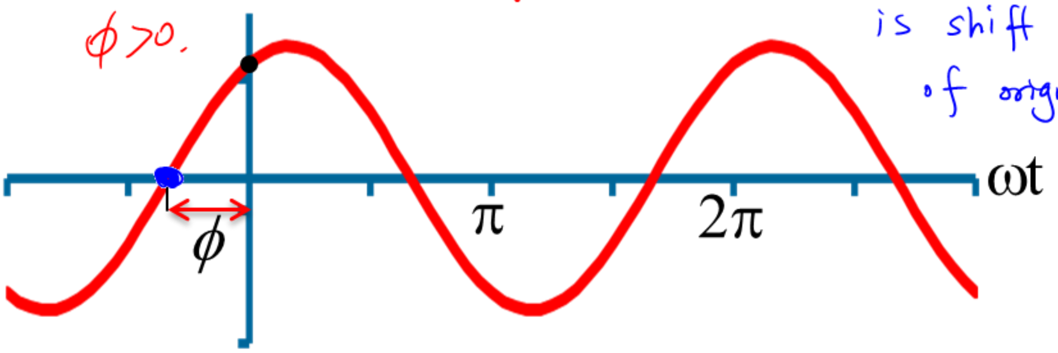
$\omega t = -\phi.$



$\omega t = -\phi < 0.$

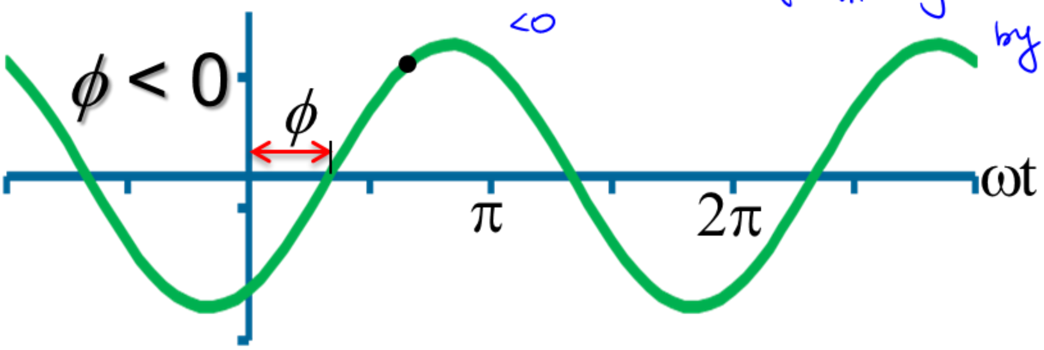
When $\phi > 0$, the signal is shift to the left of origin by ϕ.

$\phi > 0.$



$\omega t = -\underset{<0}{\phi} > 0.$

the signal is shifted to the right of origin by ϕ.



Trigonometric identities

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

$$\sin(\omega t \pm 180°) = \sin \omega t \cos 180° + \cos \omega t \underbrace{\sin 180°}_{0}$$

$$= -\sin \omega t$$

$$\cos(\omega t \pm 180°) = \cos \omega t \cos 180° - \sin \omega t \sin 180°$$

π

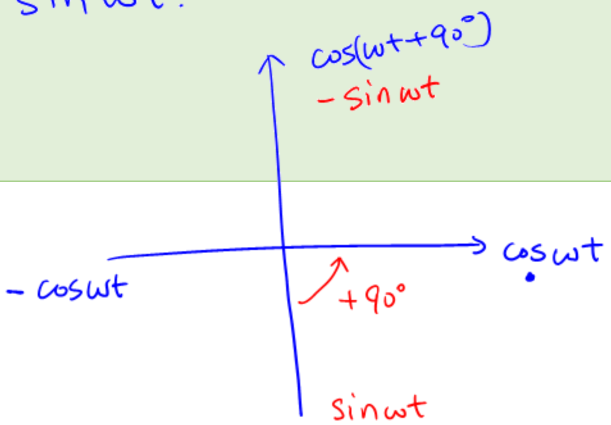
$$= -\cos \omega t.$$

$$\sin(\omega t \pm 90°) = \sin \omega t \underbrace{\cos 90°}_{0} + \cos \omega t \underbrace{\sin 90°}_{1}$$

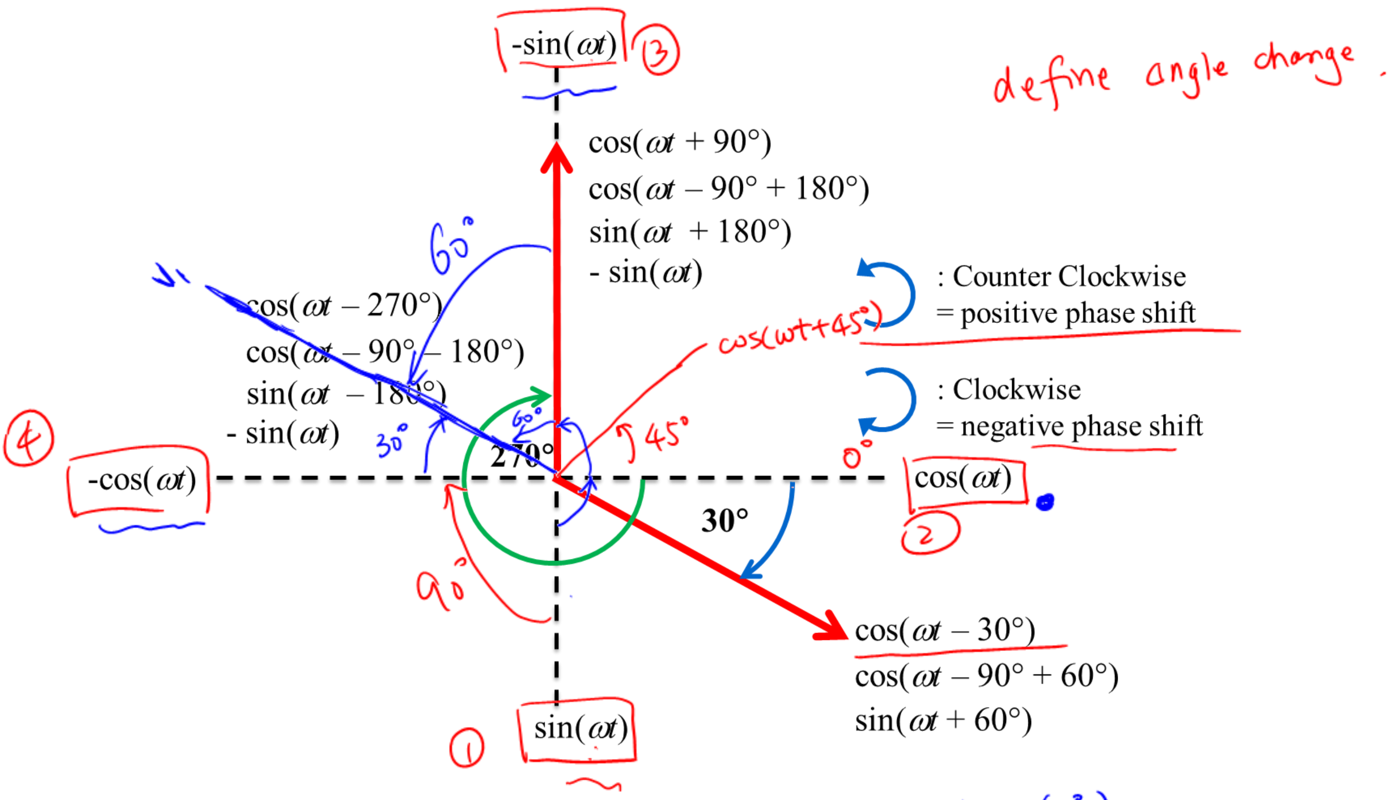
$$= \cos \omega t$$

$$\cos(\omega t \pm 90°) = \cos \omega t \cos 90° - \sin \omega t \sin 90°$$

$$= -\sin \omega t.$$



Sinusoid transform (graphic approach)



$V_1 = -3\cos(\omega t - 30^\circ)$,

(a) $V_1 = -3\sin(\omega t + 60^\circ)$

(b) $V_1 = 3\cos(\omega t + 150^\circ)$.

(c) $V_1 = 3\sin(\omega t - 120^\circ)$.

$= 3\sin(\omega t + 240^\circ)$.

$240^\circ - 360^\circ$.

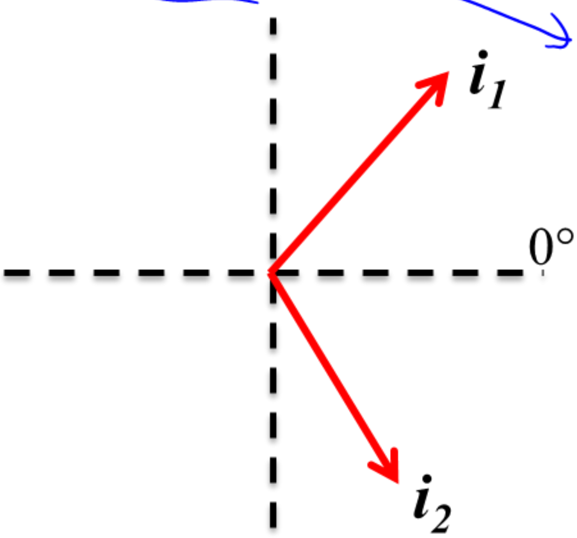
$-\cos$

$-\sin$

$\cos$

$\sin$.

Sinusoid: Lead and lag



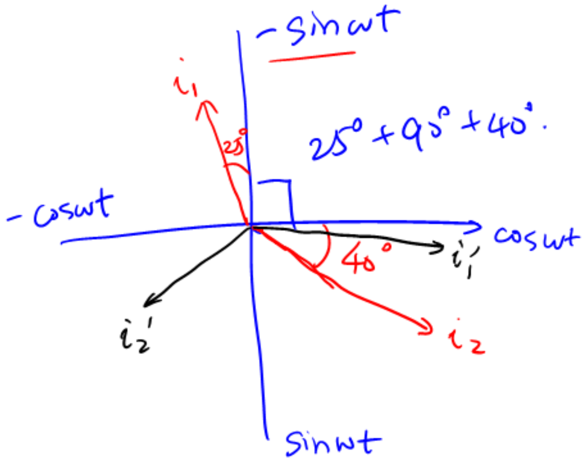
$i_1 = I_m \cos(\omega t + \phi)$.
I_m to be positive.
$\phi \in [-180°, 180°)$.
must be cosine fontion

1. • Transform both sinusoid into cosine functions.
2. • Both vectors must start in bottom two quadrants. III, IV, if not, we must rotate both vectors by the same angle to force them to be in quadrants III, IV.
3. • Rotate both vectors counter-clockwise, the one passes 0° first leads.

Example:

$$i_1 = -4sin(377t + 25°)$$

$$i_2 = 5\,cos(377t - 40°)$$



① $i_1 = 4\cos(377t + 115°)$.

② Rotate both signal by $-120°$.
$i_1' = 4\cos(377t - 5°)$. ✓
$i_2' = 5\cos(377t - 160°)$

③ rotate i_1', i_2'
i_1' passes 0° first.
i_1 leads i_2 by 155°.

Example 1: Sinusoid

For $f(t) = 5\,sin\,(4\pi t - 60°)$, determine:

Amplitude	5
Phase	$-60°$
Angular frequency	$\omega = 4\pi$, $T = \frac{2\pi}{\omega} = \frac{2\pi}{4\pi} = 0.5$ [s].
Frequency	$f = \frac{1}{T} = 2$ Hz.
Period	$T = 0.5$ [s].

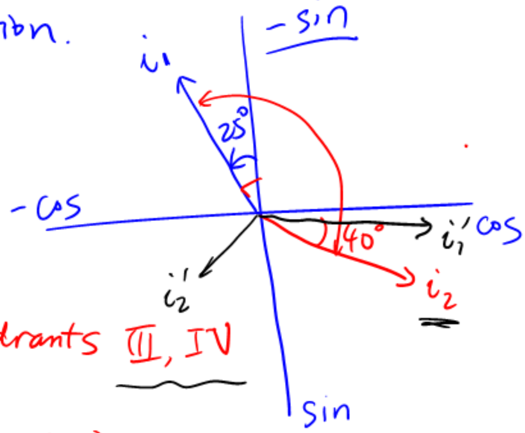
Example 2: Lead-lag $\theta = 25^\circ + 90^\circ + 40^\circ = 155^\circ$.

Determine the phase angle between $i_1 = -4sin(377t + 25°)$, and $i_2 = 5\,cos(377t - 40°)$, then comment on which signal is leading on the complex domain.

① both must be in cosine function.

$i_1 = 4\cos(377t + 115^\circ)$

$i_2 = 5\cos(377t - 40^\circ)$.



② force both i_1, i_2 into quadrants III, IV

Let's rotate both i_1, i_2 by 125°

$i_1' = 4\cos(\overset{377}{\omega} t + 115^\circ - 125^\circ) = 4\cos(\omega t - 10^\circ)$

$i_2' = 5\cos(\omega t - 40^\circ - \underset{128^\circ}{125^\circ}) = 5\cos(\omega t - 165^\circ)$.

Don't rotate it with an angle $> 140^\circ$.

③ Rotate i_1', i_2', i_1' cuts (passes) 0° first.

i_1 leads i_2. by 155°.

i_2 lags i_1 by 155°.